

Ketan Mhaske

Linux Systems Engineer | C/C++ Developer | Systems & Security Programming

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Professional Summary

Linux Systems Engineer with 2.7+ years of experience developing software using C/C++, Linux internals, networking, and security technologies. Experienced in building Linux daemons, packet inspection engines, authentication systems, cloud synchronization utilities, and enterprise security solutions. Strong expertise in debugging production issues, troubleshooting system-level problems, and building reliable, high-performance Linux applications.

Technical Skills

Programming Languages: C, C++, STL, SQL, Bash

Core Computer Science: Data Structures, Algorithms, Object-Oriented Programming

Operating Systems: Linux, Ubuntu, CentOS, RHEL

System Concepts: Memory Management, Shared Memory, IPC, Process Synchronization, Event-Driven Programming

Linux & System Programming: Linux Internals, Processes, Threads, Signal Handling, Daemon Services, epoll, File I/O, System Calls

Networking: TCP/IP, UDP, Socket Programming, TLS/SNI Parsing, libpcap, Netfilter

Concurrency: Multithreading, Synchronization Mechanisms, Thread-Safe Design

Libraries & Build Tools: Make, CMake, SQLite3, spdlog

Security: PAM, Two-Factor Authentication (2FA), Device Control, Endpoint Security

Debugging & Tools: GDB, Valgrind, Git, Vim, Nano

Professional Experience

System Analyst (Associate) – NSE Client Project

Mar 2026 – Present

NuSummit Technologies

Mumbai, India

- Contributing to Linux-based trading infrastructure supporting NSE International Exchange (Gift Nifty), covering contract processing, backend utilities, and production support.
- Developed shell automation utilities for deployment workflows, environment setup, and operational task optimization.
- Performed log analysis, system monitoring, and incident troubleshooting to ensure application reliability and availability.
- Collaborated across teams to validate deployments and trace end-to-end trading infrastructure dependencies.

Software Developer – Linux Security & Systems Engineering

Nov 2023 – Feb 2026

MicroWorld Software Services Pvt. Ltd.

Mumbai, India

- Developed and maintained enterprise Linux endpoint security solutions covering device control, authentication enforcement, network monitoring, and data protection.
- Designed Linux daemon services using C/C++, multithreading, IPC, socket programming, and Linux kernel interfaces.
- Built system-level components with libudev, Netfilter, and PAM for real-time policy enforcement and access control.
- Implemented asynchronous event handling and optimized long-running services for performance and reliability.
- Investigated and resolved complex production issues using GDB and Valgrind, collaborating with QA and security teams.

Key Projects

USB Storage Encryption Kernel Module

C, Linux Kernel, BIO, VFS

- Developed a Linux kernel module to transparently encrypt/decrypt USB storage I/O by intercepting the BIO and VFS layers.
- Implemented a file-system agnostic encryption mechanism while preserving compatibility with existing applications.
- Maintained compatibility across Linux kernel versions 5.4 through 6.12+.

Device Control Daemon (Linux)

C++, libudev, Netfilter

- Built a Linux daemon to monitor and control USB, CD-ROM, Bluetooth, and Wi-Fi devices via policy-based rules.
- Implemented real-time device detection using libudev and enforced system-wide access restrictions.
- Integrated rule validation and event handling for enterprise endpoint security use cases.

Linux Two-Factor Authentication (2FA)

C++, PAM

- Implemented PAM-based OTP authentication for SSH and local Linux login systems.
- Supported per-user and system-wide policy configurations.
- Integrated time-based OTP verification without disrupting existing authentication workflows.

Website Blocking & Packet Inspection Engine

C++, libpcap, TLS

- Built a packet inspection engine to analyze TLS handshakes and extract SNI information.
- Mapped network flows to originating processes for application-level access control.
- Implemented metadata-based HTTPS domain blocking without decrypting encrypted payloads.

Multi-Cloud User Data Sync & Backup Library

C/C++, libcurl, OAuth2

- Developed a synchronization engine integrating Google Drive, Dropbox, and OneDrive APIs.
- Implemented OAuth2 authentication, token refresh, and resumable file transfers.
- Designed failure recovery mechanisms to improve reliability for large file synchronization operations.
- Implemented automatic synchronization logic to maintain consistency between local and cloud storage.

Education

B.Tech in Computer Engineering

2019 – 2023

Deogiri Institute of Engineering and Management Studies, Aurangabad

CGPA: 7.5

HSC (Science)

2018 – 2019

Samarth Junior College of Arts & Science

Percentage: 69%

SSC

2016 – 2017

Swami Vivekanand Academy School

Percentage: 78.20%